DESIGN & VALIDATION OF A TENSILE TESTING JIG

Documentation

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Background

Composite materials are widely used in Formula SAE (FSAE) vehicle structures due to their high specific strength and stiffness. In applications where mechanical fasteners or inserts are bonded into composite laminates, the ability of the structure to transfer loads normal to the laminate surface is a critical design consideration. Unlike metallic structures, composites are particularly sensitive to through-thickness loading and localised stress concentrations, making experimental validation essential.

To ensure structural integrity and compliance with FSAE regulations, composite inserts must demonstrate sufficient resistance to tensile pull-out loads. The composite pull-out test evaluates the load-bearing capacity of an insert bonded or embedded within a laminate when subjected to axial tension. This test provides validation of the laminate layup, insert design, bonding method, and manufacturing quality. This report presents the composite pull-out testing conducted in accordance with FSAE requirements to verify insert strength and demonstrate regulatory compliance.

Project Overview

The objective of this project is to design and validate a structural pull-out test fixture for the mechanical characterisation of composite panels used in the Formula Student Structural Equivalency Spreadsheet (SES).

The fixture was required to exhibit negligible structural compliance under an applied pull-out load of 15 kN, ensuring that the measured force–displacement response is governed by deformation of the composite specimen rather than elastic deflection of the test apparatus.

This design requirement is critical for obtaining representative experimental data to support SES compliance, material characterisation, and correlation between physical testing and analytical predictions.

Objectives

- Design a testing jig for a composite flat panel that can be fitted into the Instron
- Perform finite element analysis to ensure the testing jig doesn't undergo any deformation under load
- Document the entire design to manufacture process and any relevant regulated to the design of the jig or testing procedure

Rules & Compliance: ISO 527-1

7 Number of Test Specimen - 7.1: A minimum of five test specimens shall be tested for each of the required directions of testing. The number of measurements may be more than five if greater precision of the mean value is required. It is possible to evaluate this by means of the confidence interval (95 % probability, see ISO 2602).

8 Conditioning: The test specimen shall be conditioned as specified in the appropriate standard for the material concerned. In the absence of this information, the most appropriate set of conditions from ISO 291 shall be selected and the conditioning time is at least 16 h, unless otherwise agreed upon by the interested parties, for example, for testing at elevated or low temperatures.

The preferred atmosphere is (23 ± 2) °C and (50 ± 10) % R.H., except when the properties of the material are known to be insensitive to moisture, in which case humidity control is unnecessary.

9.1 Test Atmosphere: Conduct the test in the same atmosphere used for conditioning the test specimen, unless otherwise agreed upon by the interested parties, for example, for testing at elevated or low temperatures.

Assumptions

- Friction between the composite component and the top plate is assumed to be negligible.
- The M8 bolt reaction forces for the structural steel plate are assumed to be representative of those for the composite plate.
- The jig's top plate is assumed to carry the full reaction forces
- The top plate and flanges were modelled as a single merged geometry to simplify the FEA and avoid explicit weld/contact modelling, assuming this provides equivalent structural rigidity to the actual fillet-welded connection.

Project

Design

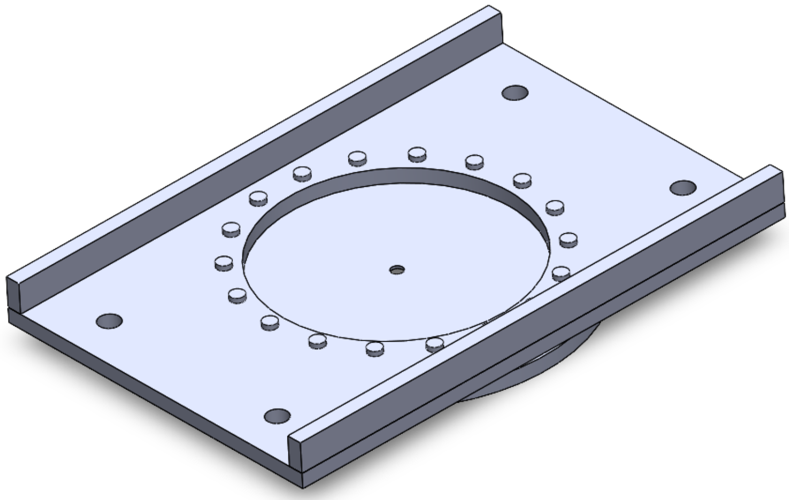


Figure 1: CAD Model of the Testing Jig (SolidWorks)

The design consists of eighteen M8 bolts configured in a circular pattern fixing a carbon plate between two 16mm (standard thickness) steel plate and four M20 threaded rods fixing the jig to the Instron. The flanges on the sides are two rectangular steel pieces that are welded on, to increase second moment of area to decrease unwanted plate deflection.

Finite Element Analysis

Approximation of Carbon Plate Forces:

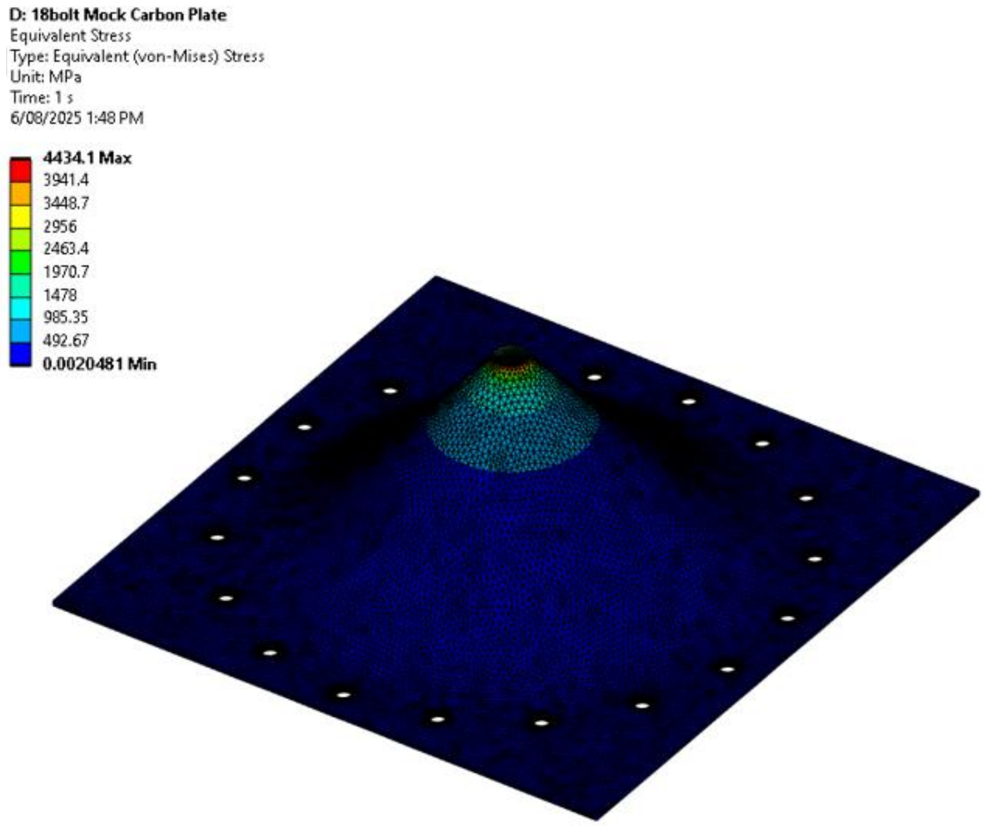


Figure 2: FEA of Mock Composite Plate

To determine the boundary conditions to be applied in the FEA of the testing jig, an initial simulation was conducted using a mock carbon plate (sheet steel). The force reactions at the M8 holes, shown in Figure 7, were obtained from this simulation. Figure 2 illustrates the resulting stress distribution across the plate, while Figure 3 shows the inverted reaction forces used to define the boundary conditions for the subsequent FEA of the testing jig.

Boundary Conditions:

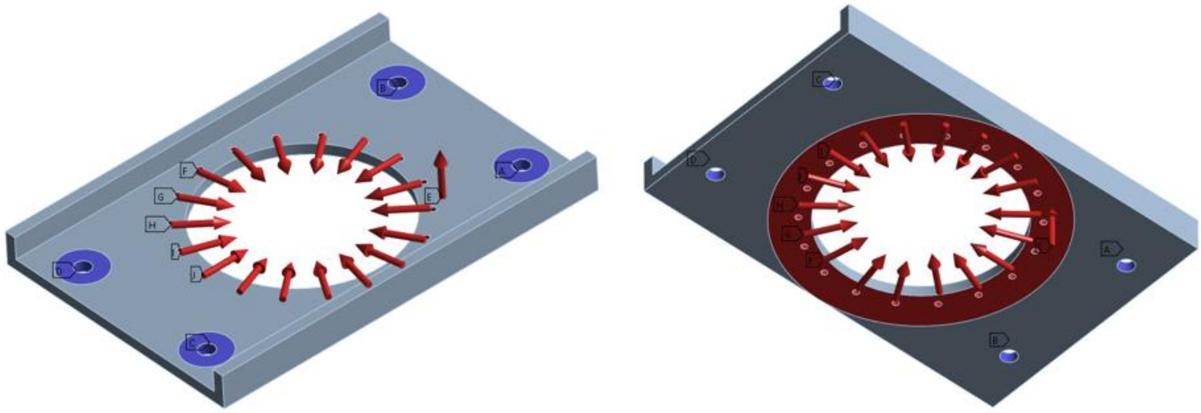


Figure 3: FEA Boundary Conditions used for Structural Simulation

To ensure the computational model accurately represents the physical behaviour of the system, appropriate boundary conditions (BC) were carefully defined and applied. The following boundary conditions were implemented, along with their corresponding justifications:

1. **Compression-only supports** were applied to the M20 rod holes to represent the contact interaction between the jig and the rods. This condition allows load transfer only when the hole walls are in compression against the rods, accurately capturing the contact stresses that arise as the jig deflects away from the rods.

2. A **frictionless support** was assigned to the washer to constrain displacement in the normal direction while permitting tangential motion. This approach simulates contact between components without introducing artificial normal deformation or frictional resistance.

3. **Reaction forces** were applied at the M8 bolt holes to counterbalance the externally applied loads acting on the mock composite plate. In addition, a **15 kN vertical load** was applied to the underside of the plate to replicate the loading conditions imposed by the Instron testing machine.

Deformation:

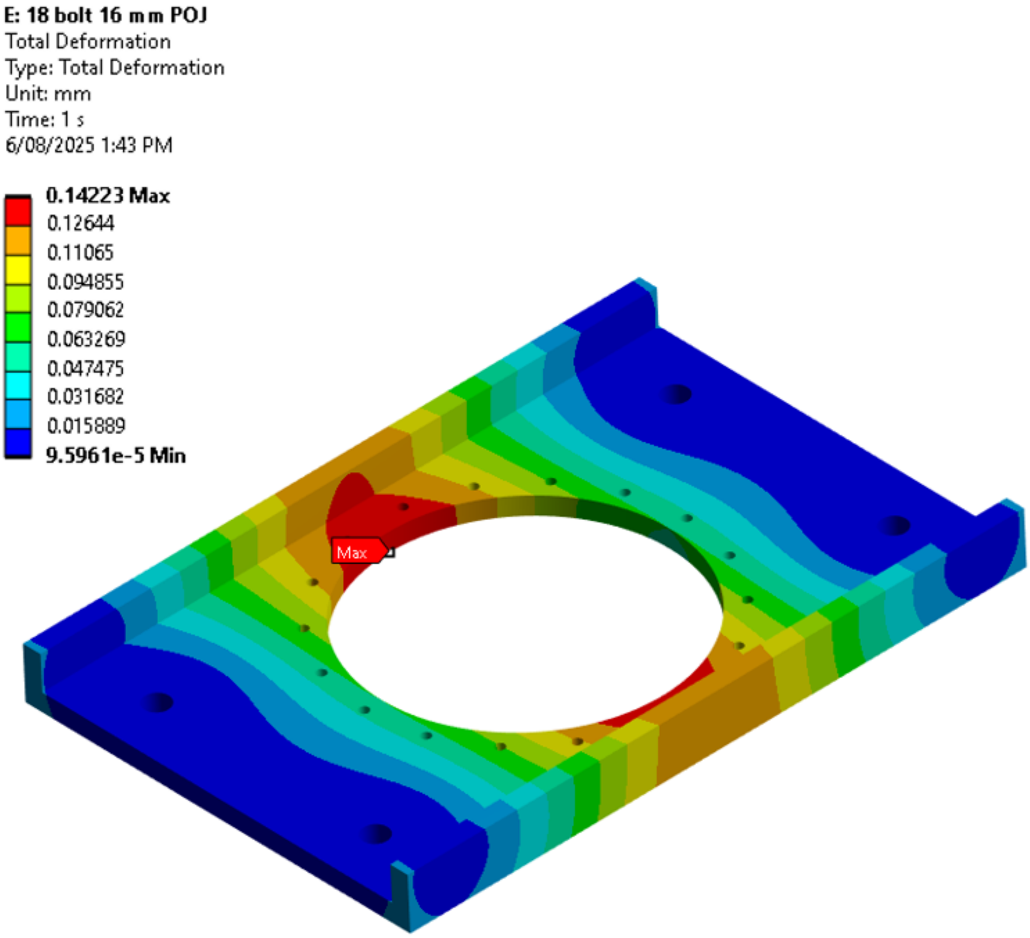


Figure 4: Deformation Contour of the Testing Jig

Figure 4 shows that the largest deformation are located in the central region of the specimen. This behaviour is expected, as the central region is the furthest from the fixture and therefore experiences the greatest bending moment.

Von Mises Stress:

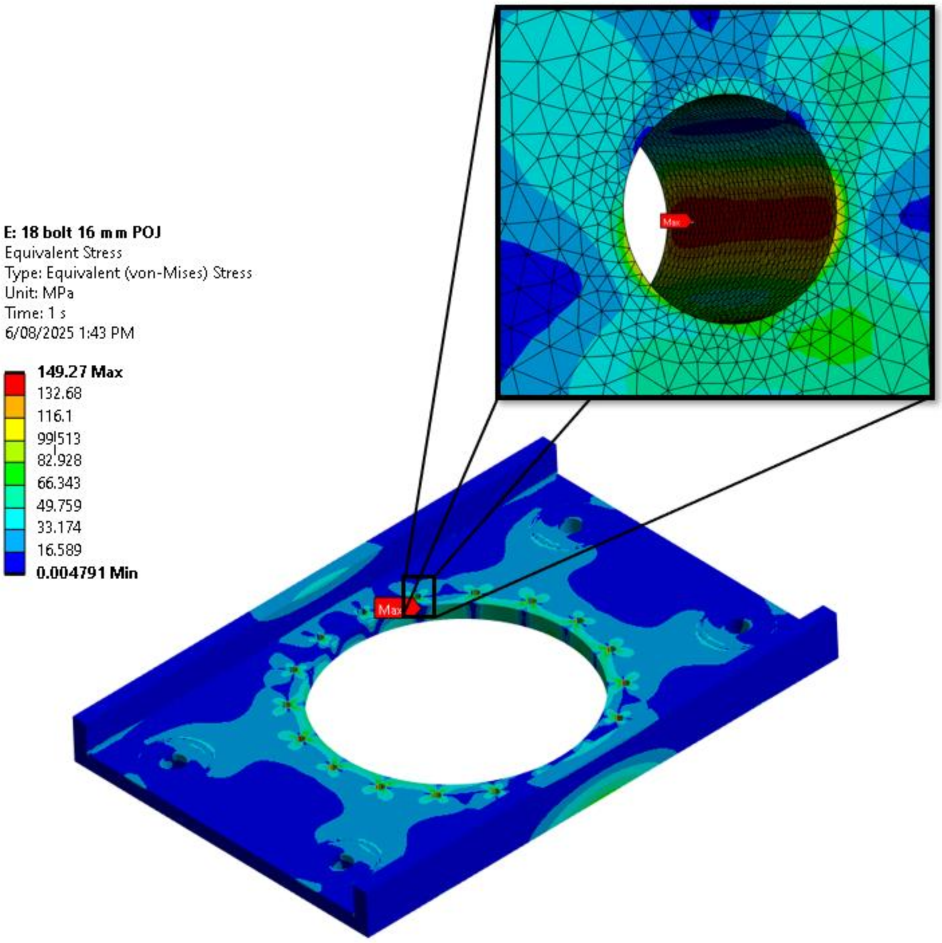


Figure 5: Stress Contour and Concentration of Testing Jig

Figure 5 illustrates the stress distribution across the entire jig. The results indicate localised stress concentrations within the M8 bolt holes. This behaviour is expected, as these regions are primarily responsible for resisting deformation caused by the pull-out force which results in the plate compression against the bolts.

Results

Following approval of the finite element analysis (FEA) model by a certified professional engineer, the testing jig was released for fabrication through the university machine shop. Upon completion, the jig was immediately commissioned for experimental validation, as shown in Figure 6. The predicted deflection from the FEA simulation was **0.14223 mm**, compared with a hand-calculated deflection of **3.52213×10^{-5} mm**, representing a difference of approximately **403,710%**.

This substantial discrepancy is primarily attributed to the simplifying assumptions used in the hand calculation, which did not account for the central hole, resulting in an overestimation of the effective cross-sectional area and, consequently, an underestimation of the predicted deflection. In contrast, the FEA model incorporated the detailed geometry and corresponding reduction in load-bearing cross-sectional area, providing a more representative prediction of the jig's structural response.

The test results demonstrated that a tensile load of **25 kN** was successfully applied through the composite specimen with no observable deformation or measurable deflection of the testing jig, supporting the structural adequacy of the design and the validity of the FEA and hand calculation predictions.

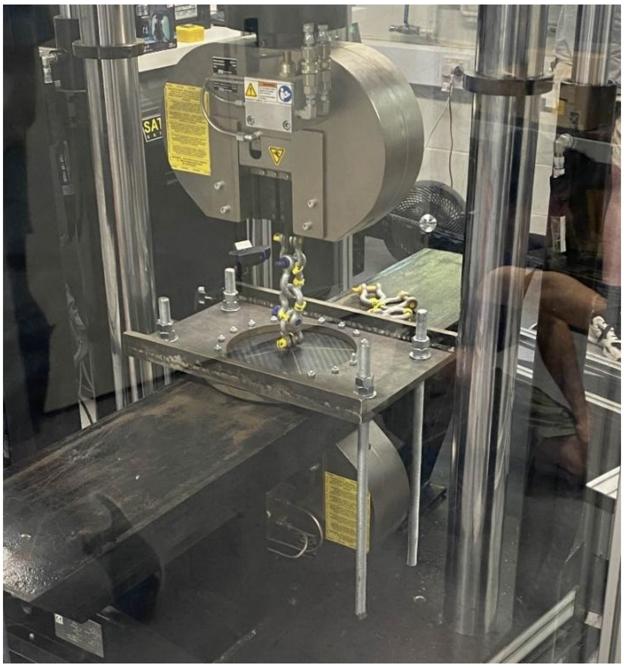


Figure 6: Composite Testing Jig in Use

Appendix

Mock Composite Plate Solution Tree

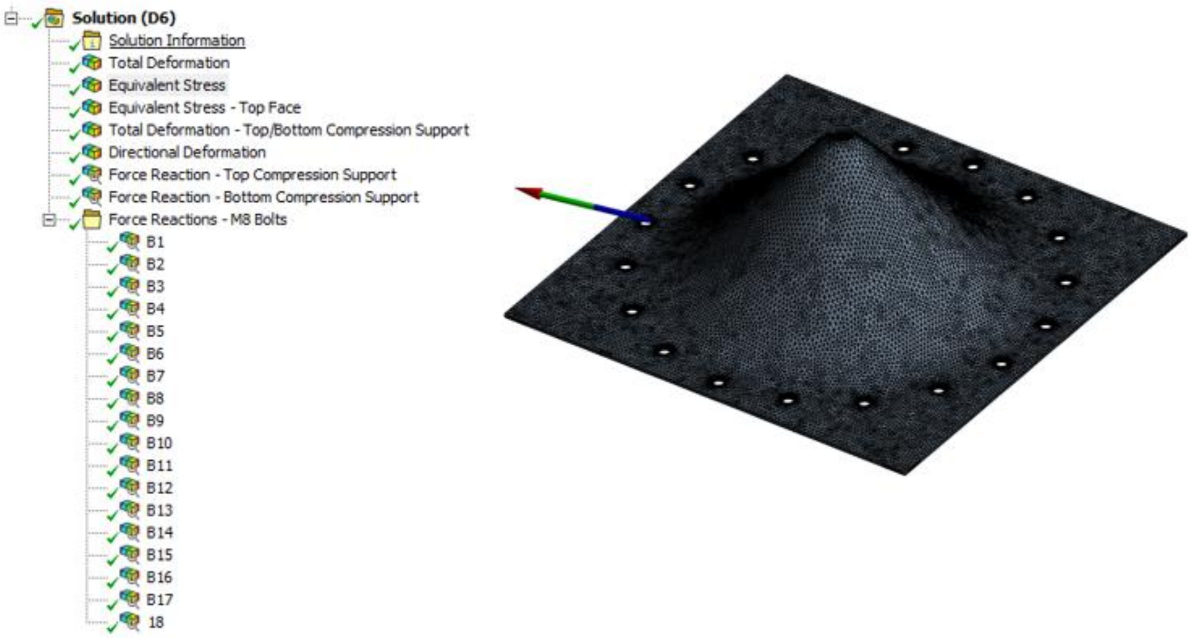
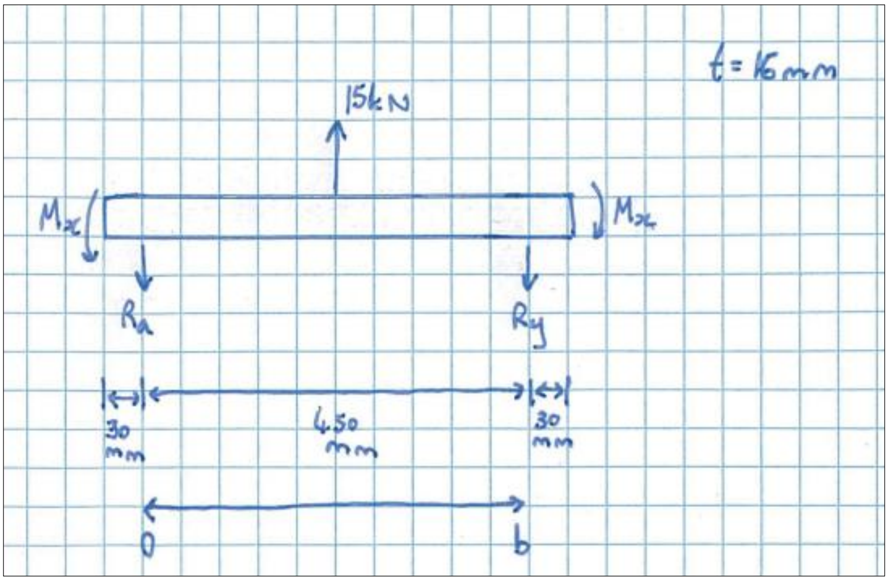


Figure 7: Finite Element Analysis for M8 Bolt Reaction Forces

Hand Calculations: Plate Deflection



Boundary Conditions

Deflection at R_a & $R_b = 0$

Therefore,

$$w(y = 0) = 0 \ \cdots [1]$$

$$w(y = b) = 0 \ \cdots [2]$$

Slope at R_a & $R_b = 0$

Therefore,

$$\frac{\partial w}{\partial y}(y = 0) = 0 \ \cdots [3]$$

$$\frac{\partial w}{\partial y}(y = b) = 0 \cdots [4]$$

Governing Equation

$$\frac{\partial^4 w}{\partial x^2} + 2 * \frac{\partial^4 w}{\partial w^2 \partial y^2} + \frac{\partial^4 w}{\partial y^4} = \frac{P}{D}$$

Since there's no change in the x plane the first two terms equate to zero therefore, the governing equation becomes.

$$\frac{\partial^4 w}{\partial y^4} = \frac{P}{D}$$

Derivations

$$\frac{\partial^3 w}{\partial y^3} = \frac{P}{D} y + C_1$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{1}{2}\frac{P}{D} y^2 + C_1 y + C_2$$

$$\frac{\partial w}{\partial y} = \frac{1}{6}\frac{P}{D} y^3 + \frac{1}{2} C_1 y^2 + C_2 y + C_3$$

$$w = \frac{1}{24}\frac{P}{D} y^4 + \frac{1}{6} C_1 y^3 + \frac{1}{2} C_2 y^2 + C_3 y + C_4$$

Substituting in boundary conditions

B.C. [1]: When $y = 0, w = 0$

$$w = \frac{1}{24}\frac{P}{D} y^4 + \frac{1}{6} C_1 y^3 + \frac{1}{2} C_2 y^2 + C_3 y + C_4$$

$$0 = \frac{1}{24}\frac{P}{D} 0^4 + \frac{1}{6} C_1 0^3 + \frac{1}{2} C_2 0^2 + C_3 0 + C_4$$

$$\therefore C_4 = 0$$

B.C. [3]: When $y = 0, \frac{\partial w}{\partial y} = 0$

$$\frac{\partial w}{\partial y} = \frac{1}{6}\frac{P}{D} y^3 + \frac{1}{2} C_1 y^2 + C_2 y + C_3$$

$$0 = \frac{1}{6}\frac{P}{D} 0^3 + \frac{1}{2} C_1 0^2 + C_2 0 + C_3$$

$$\therefore C_3 = 0$$

B.C. [4]: When $y = b, \frac{\partial w}{\partial y} = 0$

$$\frac{\partial w}{\partial y} = \frac{1}{6}\frac{P}{D} y^3 + \frac{1}{2} C_1 y^2 + C_2 y$$

$$0 = \frac{1}{6}\frac{P}{D} b^3 + \frac{1}{2} C_1 b^2 + C_2 b$$

$$C_2 = -\frac{1}{6}\frac{P}{D} b^2 - \frac{1}{2} C_1 b$$

B.C. [2]: When $y = b, w = 0$

$$w = \frac{1}{24}\frac{P}{D} y^4 + \frac{1}{6} C_1 y^3 + \frac{1}{2} C_2 y^2$$

$$0 = \frac{1}{24}\frac{P}{D}b^4 + \frac{1}{6}C_1 b^3 + \frac{1}{2}C_2 b^2$$

Substitute in C_2 from previous equation to get C_1

$$0 = \frac{1}{24}\frac{P}{D}b^4 + \frac{1}{6}C_1 b^3 + \frac{1}{2}\left(-\frac{1}{6}\frac{P}{D}b^2 - \frac{1}{2}C_1 b\right)b^2$$

$$C_1 = -\frac{1}{2}\frac{P}{D}b$$

Substitute C_1 into C_2

$$C_2 = -\frac{1}{6}\frac{P}{D}b^2 - \frac{1}{2}\left(-\frac{1}{2}\frac{P}{D}b\right)b$$

$$C_2 = \frac{1}{12}\frac{P}{D}b^2$$

Substitute C_1, C_2, C_3 & C_4 into w

$$w = \frac{1}{24}\frac{P}{D}y^4 + \frac{1}{6}C_1 y^3 + \frac{1}{2}C_2 y^2 + C_3 y + C_4$$

$$w = \frac{1}{24}\frac{P}{D}y^4 + \frac{1}{6}\left(-\frac{1}{2}\frac{P}{D}b\right)y^3 + \frac{1}{2}\left(\frac{1}{12}\frac{P}{D}b^2\right)y^2 + 0 * y + 0$$

$$w = \frac{1}{24}\frac{P}{D}(y^4 - 2by^3 + b^2y^2)$$

Flex Rigidity (D)

$$D = \frac{Et^3}{12(1 - v^2)}$$

Where $E = 200\ GPa$ and Poisson's ratio v is assumed to be 0.3 for steel

$$D = \frac{(200 * 10^9) * 0.016^3}{12(1 - 0.3^2)}$$

$$D = 75018.52$$

Deflection (w)

$$w = \frac{1}{24} * \frac{15 * 10^3}{75018.52} * (y^4 - 2y^3 * 0.51 + y^2 * 0.51^2)$$

$$w = (8.33 * 10^{-3}) * (y^4 - 1.02y^3 + 0.2601y^2)$$

Max Deflection is at the mid-point $[y = 0.51/2]$

$$w = (8.33 * 10^{-3}) * (0.255^4 - 1.02 * 0.255^3 + 0.2601 * 0.255^2)$$

$$w = 3.52213 * 10^{-5}\, mm$$

Hand Calculations: Rod Moments

Moment at Fixed Ends (M20 Rods)

When $y = 0$

$$\frac{\partial^2 w}{\partial y^2} = \frac{1}{2}\frac{P}{D}0^2 + \left(-\frac{1}{2}\frac{P}{D}b\right) * 0 + \left(\frac{1}{12}\frac{P}{D}b^2\right)$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{1}{12}\frac{Pb^2}{D}$$

When $y = b$

$$\frac{\partial^2 w}{\partial y^2} = \frac{1}{2}\frac{P}{D}b^2 + \left(-\frac{1}{2}\frac{P}{D}b\right) * b + \left(\frac{1}{12}\frac{P}{D}b^2\right)$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{1}{12}\frac{Pb^2}{D}$$

M_x Moment:

$$M_x = -D\left(\frac{\partial^2 w}{\partial x^2} + v\frac{\partial^2 w}{\partial y^2}\right)$$

$$M_x = -D\left(\frac{\partial^2 w}{\partial x^2} + v\left(\frac{1}{12}\frac{Pb^2}{D}\right)\right)$$

Where $\frac{\partial^2 w}{\partial x^2} = 0$ and D cancels out

$$M_x = -\frac{1}{40}Pb^2$$

Substitute $P = 15\, kN$ & $b = 450\, mm$

$$M_x = -\frac{1}{40} * (15 * 10^3) * (0.45)^2 = -84.375\, Nm$$

M_y Moment:

$$M_y = -D\left(\frac{\partial^2 w}{\partial y^2} + v\frac{\partial^2 w}{\partial x^2}\right)$$

$$M_x = -D\left(\frac{1}{12}\frac{Pb^2}{D} + v\left(\frac{\partial^2 w}{\partial x^2}\right)\right)$$

Where $\frac{\partial^2 w}{\partial x^2} = 0$ and D cancels out

$$M_y = -\frac{1}{12}Pb^2$$

Substitute $P = 15\ kN$ & $b = 450\ mm$

$$M_y = -\frac{1}{12}Pb^2$$

$$M_y = -\frac{1}{12} * (15 * 10^3) * (0.45)^2 = -253.125\ Nm$$